

Atomic Josephson parametric amplifier

Vijay Pal Singh^{1,*}, Luigi Amico^{1,2,3}, and Ludwig Mathey^{4,5}¹Quantum Research Centre, Technology Innovation Institute, Abu Dhabi, United Arab Emirates²Dipartimento di Fisica e Astronomia, Università di Catania, Via S. Sofia 64, 95123 Catania, Italy³INFN-Sezione di Catania, Via S. Sofia 64, 95127 Catania, Italy⁴Zentrum für Optische Quantentechnologien and Institut für Quantenphysik, Universität Hamburg, 22761 Hamburg, Germany⁵The Hamburg Centre for Ultrafast Imaging, Luruper Chaussee 149, 22761 Hamburg, Germany

(Received 2 April 2025; accepted 11 February 2026; published 20 March 2026)

We study the dynamics of a driven atomic Josephson junction that we propose as a parametric amplifier. By periodically modulating the position of the barrier, we induce a small current across the junction, serving as our input signal. The pump field is implemented by modulating the barrier height at twice the Josephson plasma frequency. The resulting dynamics exhibit parametric amplification of the signal through nonlinear mixing between the signal and pump fields, which is encoded in a specific microscopic pattern of density waves and phase excitations that can be addressed within the experimental cold-atom capabilities. This work paves the way for tunable amplifiers in atomtronic circuits, with potential applications in several fields including precision measurements and quantum information processing. At the same time, our analysis provides the microscopic explanation of the general notion of parametric amplification occurring in nonlinear coherent devices.

DOI: 10.1103/PhysRevResearch.8.L012067

Parametric amplification is a fundamental concept in nonlinear physics: When the system's natural frequency ω_0 is periodically modulated in time with frequency ω_d (the pump), a nearly resonant condition can be achieved for $\omega_d \approx 2\omega_0$; in this case, an energy transfer from the pump to the system is induced [1]. This idea was implemented for classical electronics [2] and then extended to quantum optics [3,4], semiconductors [5], and superconducting materials [6–10]. Parametric amplifiers are extensively used in important technological sectors as radio astronomy [11], communications [12], radar systems [13], spectroscopy [14], and sensing [15]. In quantum technology, parametric amplifiers have become relevant for many applications ranging from the generation of squeezed states [6,7,16], entanglement and nonclassical states [17,18], and quantum-limited amplification [19] to quantum computing [20], quantum sensing, and metrology [21,22].

Superconducting Josephson junctions (JJs) provide nonlinear circuits that have been thoroughly employed to achieve a parametric amplification. These devices amplify weak microwave signals with minimal added noise thus generating high-quality amplification. When we apply a strong pump field with frequency ω_d at twice the Josephson plasma frequency ω_0 , i.e., $I_{\text{pump}}(t) \sim \sin(2\omega_0 t)$, and a weak signal field $I_{\text{signal}}(t) \sim \cos(\omega_s t)$, with frequency $\omega_s \approx \omega_0$, the nonlinear inductance of the Josephson junction produces parametric

amplification through a nonlinear mixing process between the signal and pump fields [23]. ω_0 is determined by $\omega_0 = \sqrt{2eI_c/(\hbar C)}$, where $2e$ is the electronic charge, I_c is the critical current, and C is the capacitance. The idler modes at frequency $\omega_l = \omega_d - \omega_s$ ensure the energy conservation. The signal mode at ω_s is amplified as the pump energy is transferred to it through the nonlinear mixing process.

Ultracold atoms have emerged as an ideal platform for implementing and exploring atomtronic analogs of superconducting circuits [24–33]. One breakthrough was the realization of JJs using weak links of ultracold-atom clouds [34–38]. This not only has enabled the study of important quantum effects, such as dc-ac Josephson effects [39–43], current-phase relation [40,44], and Josephson-tunneling enhanced superfluidity [45], with remarkable accuracy and control of the physical conditions, but has also laid the foundation for the development of atom-based devices, such as atomic quantum interference devices [28–30,46,47]. Theoretical studies have examined Josephson effects [48–50], the implementation of universal set of logic gates [51], atomic conductivity in pump-probe setting [52], or driven dynamics [49,53–56]. Recently, quantized plateaus of chemical potential were predicted in driven atomic JJs [57], which are the analog of Shapiro steps in superconducting JJs [58,59]. Atomic Shapiro steps were measured in Refs. [60,61], serving as a basis for a pressure standard in atomic condensates.

With remarkable progress in the field of ultracold atoms, driven matter-wave circuits are now well within experimental reach. Here, we propose the implementation of a Josephson parametric amplifier (JPA) using ultracold atoms. Our protocol relies on an atomic JJ in which we periodically modulate both the position and height of the tunnel barrier [see Figs. 1(a) and 1(b)]. This can be realized in ultracold-atom ex-

*Contact author: vijay.singh@tii.ae

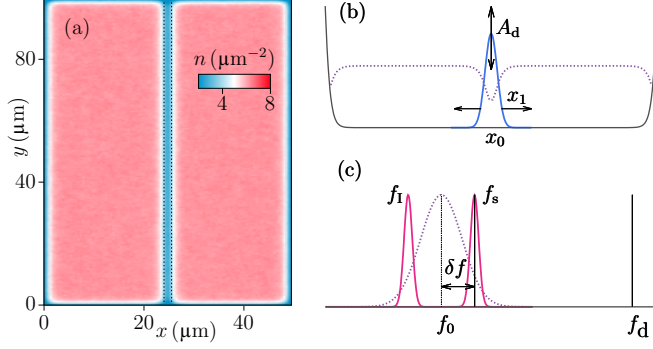


FIG. 1. Atomic JPA. (a) Simulation of a Josephson junction, which consists of two two-dimensional clouds separated by a Gaussian tunnel barrier of height V_0 and width w (indicated by two dotted vertical lines). We use $V_0/\mu = 1.5$ and $w/\xi = 1.1$, where μ is the mean-field energy and ξ is the healing length. (b) Sketch of the JPA protocol. We periodically modulate the barrier position using $x(t) = x_0 + x_1 \sin(2\pi f_s t)$, where x_1 is the amplitude and f_s is the frequency. For the pump we modulate the barrier height, i.e., $V(t) = V_0(1 + A_d \cos(2\pi f_d t))$, where A_d is the amplitude and f_d is the frequency. (c) Sketch of the operation of a JPA. We modulate the barrier height near twice the plasma frequency f_0 of the JJ. The signal mode at $f_d/2 + \delta f$ is amplified and an idler mode at frequency $f_d/2 - \delta f$ is created, with δf being the tuning parameter.

periments using digital micromirror devices [60,61]. In our system, the modulation of the barrier position serves as the input current signal, while the modulation of the barrier height acts as a pump field. By tuning the pump frequency to twice the Josephson plasma frequency, we induce nonlinear mixing between the pump and signal modes [see Fig. 1(c)]. A strong pump field drives the system into a regime where parametric amplification occurs. Our results are obtained using classical-field dynamics, which include fluctuating bosonic fields beyond the mean-field description [62–65]. Additionally, the system dynamics are well captured by a driven circuit model.

System and protocol. We consider a homogeneous cloud of bosons confined in a box of dimensions $L_x \times L_y$. We study the dynamics of this system using classical-field dynamics within the truncated Wigner approximation [62–65]. The system is described by the Hamiltonian

$$\hat{H}_0 = \int d\mathbf{r} \left[\frac{\hbar^2}{2m} \nabla \hat{\psi}^\dagger(\mathbf{r}) \cdot \nabla \hat{\psi}(\mathbf{r}) + \frac{g}{2} \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}) \hat{\psi}(\mathbf{r}) \right]. \quad (1)$$

$\hat{\psi}$ ($\hat{\psi}^\dagger$) is the bosonic annihilation (creation) operator. The interaction $g = \tilde{g}\hbar^2/m$ is given in terms of the dimensionless parameter $\tilde{g} = \sqrt{8\pi}a_s/\ell_z$, where m is the mass, a_s is the s -wave scattering length, and $\ell_z = \sqrt{\hbar/(m\omega_z)}$ is the harmonic oscillator length in the transverse direction. In the limit of a large atom number N , we employ the classical-field representation, where the operators $\hat{\psi}$ in Eq. (1) and the equations of motion are replaced by complex numbers ψ . The initial states $\psi(\mathbf{r}, t=0)$ are generated in a grand canonical ensemble with chemical potential μ and temperature T via a classical

Metropolis algorithm. This distribution captures fluctuations of $\psi(\mathbf{r}, t=0)$ around its mean field value. Finally, each initial state is evolved according to the classical equations of motion

$$i\hbar\dot{\psi}(\mathbf{r}, t) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}, t) + g|\psi|^2 \right) \psi(\mathbf{r}, t), \quad (2)$$

which include the barrier potential given by $V(\mathbf{r}, t) = V_0(t) \exp[-2(x - x(t))^2/w^2]$, where $V_0(t)$, w , and $x(t)$ are the barrier's strength, width, and location, respectively. We perform numerical simulations by discretizing space on a lattice of size $N_x \times N_y = 100 \times 200$ with the discretization length $l = 0.5 \mu\text{m}$. To illustrate the key idea, we focus on a concrete realization using $^6\text{Li}_2$ molecules, though our protocol is applicable to any cold-atom degenerate gas. We set the density to $n \approx 5.6 \mu\text{m}^{-2}$, interaction parameter $\tilde{g} = 0.1$, and temperature ratio $T/T_0 = 0.06$, with system dimensions $L_x \times L_y = 50 \times 100 \mu\text{m}^2$. The critical temperature T_0 is estimated by $T_0 = 2\pi n \hbar^2 / (mk_B \mathcal{D}_c)$, where $\mathcal{D}_c = \ln(380/\tilde{g})$ is the critical phase-space density [66,67]. We use $w/\xi = 1.1$ and $V_0/\mu \equiv \tilde{V}_0 = 1.5$, where $\xi = \hbar/\sqrt{2mgn}$ is the healing length and $\mu = gn$ is the mean-field energy. To create the weak link at $x(t) = x_0 = L_x/2$, we linearly ramp up V_0 over 200 ms and then wait for 50 ms. The density distribution $n(x, y) = |\psi(\mathbf{r}, t)|^2$ is shown in Fig. 1(a), which is averaged over the initial ensemble. For the signal, we periodically modulate the barrier position using $x(t) = x_0 + x_1 \sin(2\pi f_s t)$, where x_1 is the amplitude and f_s is the frequency [see Fig. 1(b)]. The resulting barrier velocity $v(t) = v_1 \cos(2\pi f_s t)$, with $v_1 = 2\pi f_s x_1$, induces an oscillating current across the junction. We calculate the atom number $N_L(t)$ ($N_R(t)$) in the left (right) reservoir to determine the imbalance $z(t) = (N_L(t) - N_R(t))/N$, where $N = (N_L + N_R)$ is the total atom number. For the spectrum analysis, we calculate the power spectrum

$$S_\omega = |z(\omega)|^2, \quad (3)$$

where $z(\omega) = (1/\sqrt{T_n}) \int dt z(t) \exp(-i\omega t)$ is the Fourier transform of $z(t)$. T_n is the sampling time for the numerical Fourier transform. The junction displays a well-defined resonance at the Josephson plasma frequency f_0 , which we determine by analyzing S_ω at varying frequency, obtaining $f_0 = 39.4 \text{ Hz}$ in the limit of $v_1 \rightarrow 0$ (see Appendix A: *Josephson plasma frequency*). For the parametric pump, we modulate the barrier height, i.e., $V(t) = V_0(1 + A_d \cos(2\pi f_d t))$, where A_d is the amplitude and f_d is the frequency, as depicted in Fig. 1(b). This alters the nonlinear inductance of the junction, which gives rise to amplification of the signal due to energy exchange between different modes. When $V(t)$ is modulated at twice the plasma frequency, the system undergoes parametric amplification, where a signal at frequency $f_d/2 + \delta f$ is amplified and an idler at frequency $f_d/2 - \delta f$ is created, with δf being the tuning parameter [see Fig. 1(c)]. In Fig. 2, we demonstrate the parametric enhancement for the signal at $f_s = 30 \text{ Hz}$ with respect to the system without pump. The system quickly reaches the steady state after 2–3 signal cycles and the enhancement persists over long times as required for robust amplification. The power spectrum exemplifies the amplification in frequency space.

Characterization of the JPA dynamics. To characterize the dynamical regimes we map out the imbalance spec-

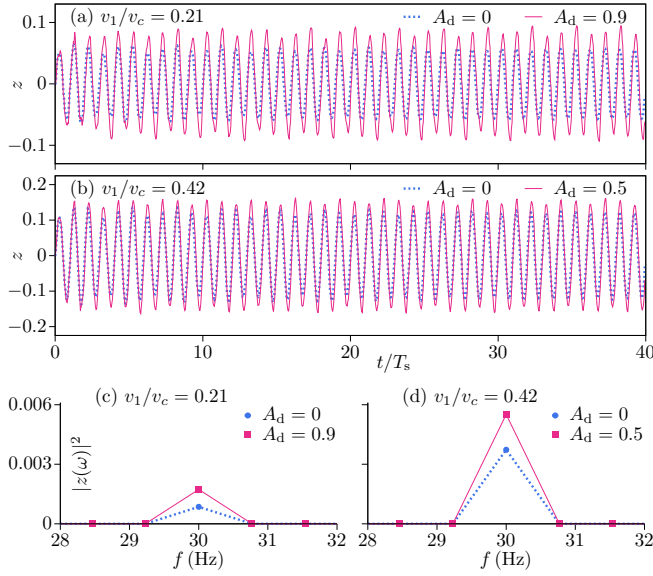


FIG. 2. JPA dynamics. Time evolution of the imbalance $z(t)$ demonstrating the amplification for the signal at frequency $f_s = 30$ Hz and two different sets of signal and pump amplitudes: (a) $v_1/v_c = 0.21$ and $A_d = 0.9$. (b) $v_1/v_c = 0.42$ and $A_d = 0.5$. v_c is the critical velocity above which the junction is resistive. The results at $A_d = 0$ represent the system without the pump. $T_s = 1/f_s$ is the signal oscillation period. (c), (d) Power spectrum of the time evolution in panels (a) and (b).

trum for signal frequencies between $f_s = 20$ and 100 Hz, which exhibits JPA resonances and other excitation peaks (see the Supplemental Material [68]). Figure 3 highlights the interplay between the signal and idler modes for the lowest JPA resonance. The mode weights are controlled by the pump amplitude. To describe the JPA dynamics, we introduce a driven RSJ (resistively shunted junction) circuit model. The barrier velocity results in a modulation of the current across the junction given by $I_1 \cos(\omega_s t)$, where $I_1 = v_1 I_c / v_c$ in which v_c is the critical velocity associated with the critical current I_c [68]. The barrier height modulation changes the tunneling energy, which can be incorporated as $I_c(1 + A_d \sin(\omega_d t))$. The Kirchhoff law of the driven RSJ circuit reads

$$I_1 \cos(\omega_s t) = I_c(1 + A_d \sin(\omega_d t)) \sin \phi - G \Delta \mu, \quad (4)$$

where $\phi = \phi_L - \phi_R$ is the phase difference across the junction and G is the conductance [68]. The Josephson relation for the phase dynamics is $\hbar \dot{\phi} = -\Delta \mu$, with $\Delta \mu$ playing the role of the voltage across the junction. We numerically solve Eq. (4) for various signal and pump parameters and obtain the time evolution $\Delta \mu(t)$. The values of $I_c = 135$ (10^3 /s) and $\hbar G = 59.8$ are determined from the simulated current-chemical potential relation [68]. With I_c and the effective charging energy $E_C = 4(\partial \mu / \partial N)$, we estimate the Josephson frequency $\omega_J = \sqrt{I_c E_C / \hbar}$ and obtain $f_J = 38$ Hz in agreement with the value of f_0 . The result of the spectrum $|\Delta \mu(\omega)|^2$ is shown in Fig. 3(d), which captures the dynamical response of the JPA. We note that the circuit model also describes higher JPA resonance and other excitation peaks well [68].

To quantify the overall amplification, we define a dimensionless gain $G_s = S_\omega(A_d)/S_\omega(0) - 1$, where $S_\omega(A_d)$

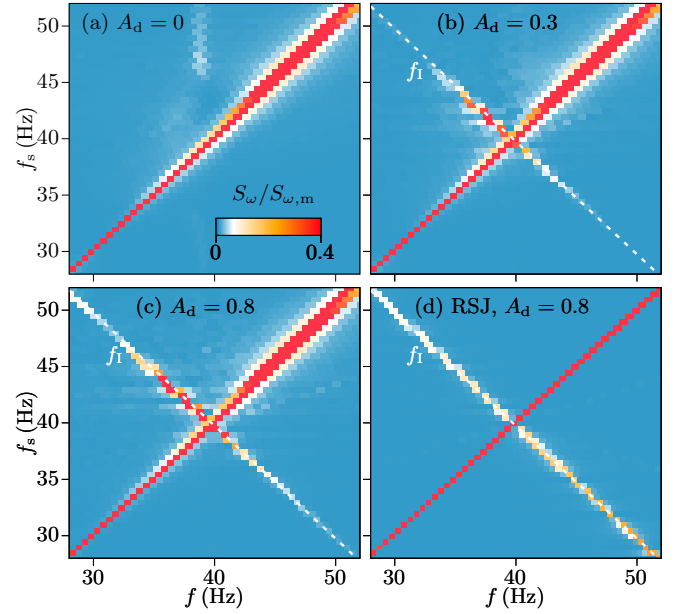


FIG. 3. Nonlinear mixing in the power spectrum $S_\omega = |z(\omega)|^2$ for signal frequencies between $f_s = 28$ and 52 Hz at $\tilde{I}_1 \equiv I_1/I_c = 0.42$. For each column, the spectrum is normalized by its maximum value $S_{\omega,m}$. (a) Without pump, the spectrum mainly displays the central peak at $f_s = f$. (b), (c) In the presence of pump there is another peak at $f_d - f_s$ (indicated by white dashed line), located at the frequency of the signal mirrored at the plasma frequency f_0 . The case of $f_s = f_0$, where signal and idler are at the same frequency, is a special case known as the degenerate case. (d) The result of the RSJ circuit model for $\tilde{I}_1 = 0.42$ and $A_d = 0.8$; see text.

represents the signal peak value and $S_\omega(0)$ corresponds to the result without pump. In Fig. 4(a), we demonstrate that G_s initially increases approximately quadratically with A_d , and then reaches a maximum for a value of A_d , that decreases with increasing $\tilde{I}_1 = I_1/I_c$. For larger values of A_d , the gain G_s decreases due to the dephasing dynamics as explained in the subsequent discussion. While the nonlinear increase is captured by the circuit model, the model fails to describe the decrease beyond the maximum, primarily visible for larger $\tilde{I}_1$. Figure 4(b) presents the signal-to-noise ratio (SNR), defined as $\bar{S}_\omega / \Delta \bar{S}_\omega$, where we compute the mean signal value $\bar{S}_\omega = \langle S_\omega(A_d) - S_\omega(0) \rangle$ and the noise $\Delta \bar{S}_\omega$ from the variance determined by $\Delta \bar{S}_\omega^2 = \langle (S_\omega(A_d) - S_\omega(0) - \bar{S}_\omega)^2 \rangle$. Here, $\langle \cdot \rangle$ denotes a statistical average over 100 samples of the ensemble. The SNR depends on both A_d and $\tilde{I}_1$, with a visible increase with increasing A_d . The maximal regime of the SNR is in the same regime as the maximal gain G_s , suggesting that this is the optimal regime of operation. In Figs. 4(c) and 4(d), the simulation results obtained at a higher temperature of $T/T_0 = 0.3$ show a noticeable reduction in overall gain, while still exhibiting a qualitatively similar trend to that observed at $T/T_0 = 0.06$. Thermal fluctuations suppress the Josephson plasma frequency f_0 and the critical current I_c , resulting in $f_0 = 35.2$ Hz and $I_c = 103$ (10^3 /s) for $T/T_0 = 0.3$ (see Appendix A: Josephson plasma frequency and Ref. [68] for details).

Figure 5(a) shows the average total number of vortices in the system, determined by calculating the local phase winding

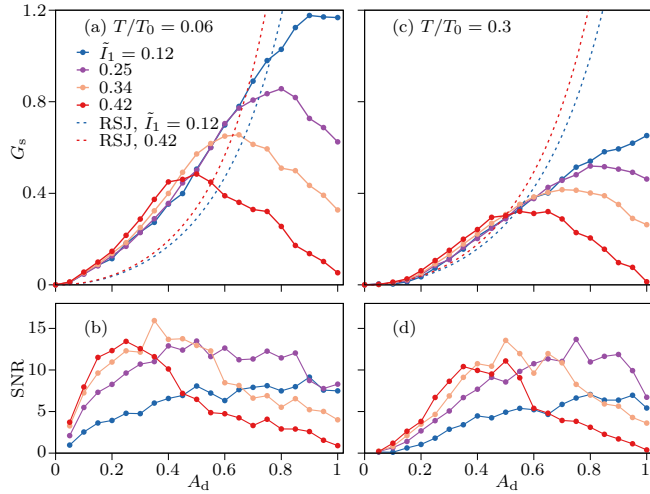


FIG. 4. JPA characteristics. (a) Dimensionless gain $G_s = S_\omega(A_d)/S_\omega(0) - 1$ as a function of A_d is shown for signal amplitudes between $\tilde{I}_1 = 0.12$ and 0.42 . The results of the RSJ model are indicated by the dashed lines. (b) SNR corresponding to the results shown in panel (a). (c), (d) Simulation and RSJ model results at temperature $T/T_0 = 0.3$, using the same relative signal frequency with respect to the Josephson plasma frequency as in panels (a) and (b); see text.

around each lattice plaquette and averaging over both the time evolution and the initial ensemble (see *Appendix B: Dephasing dynamics* for details). The increase in vortex number coincides with the slowdown and eventual reduction of the gain observed in Fig. 4(a), confirming that vortex excitations are

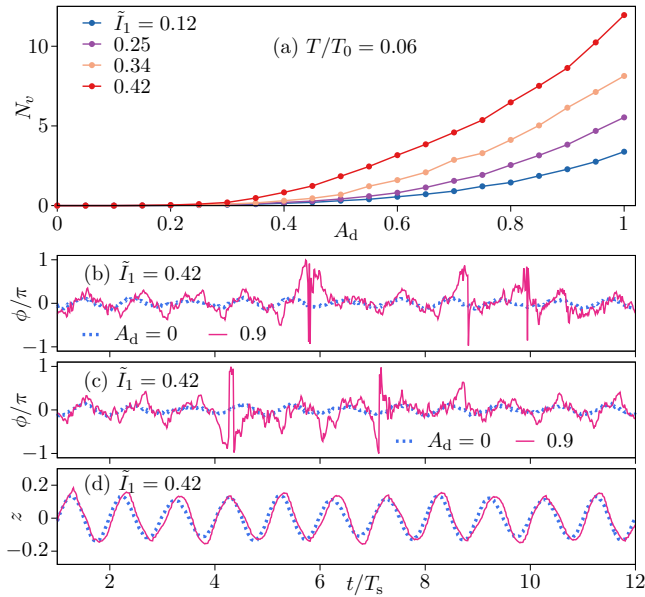


FIG. 5. Dephasing dynamics. (a) Average vortex number N_v as a function of A_d for the parameters used in Fig. 4(a). (b), (c) Time evolution of the junction phase $\phi(t)$ for two different samples at $\tilde{I}_1 = 0.42$ with (continuous line) and without pump (dashed line). (d) Imbalance $z(t)$, averaged over many samples, corresponding to the parameters in panels (b) and (c).

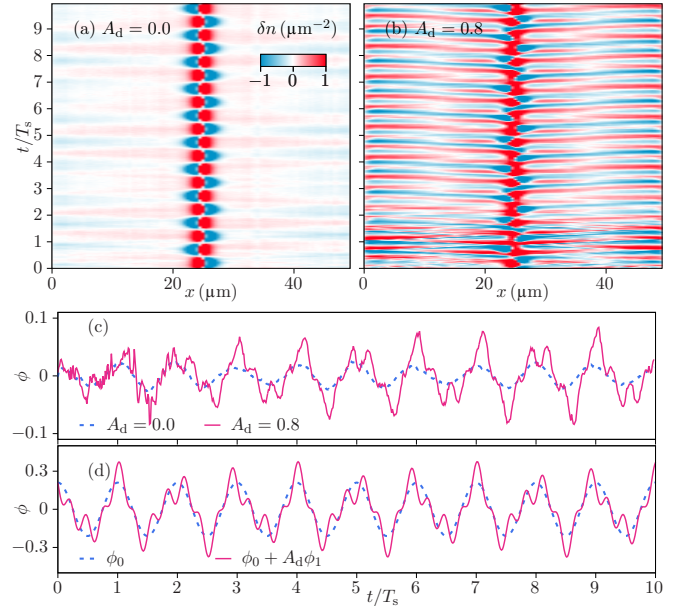


FIG. 6. Underlying nonlinear mechanism. (a) Time evolution of the density $\delta n(x, t) = n(\mathbf{r}, t) - n(\mathbf{r}, 0)$, averaged over the y direction and the ensemble, is shown for $\tilde{I}_1 = 0.21$ without the pump. (b) Dynamics in the presence of pump with $A_d = 0.8$. (c) Ensemble averaged phase difference ϕ at the junction for the cases in panels (a) and (b). (d) The results of zeroth- and first-order expansion of Eq. (4) without the resistive term; see text.

the primary dissipation mechanism limiting the amplification. During time evolution, the junction phase undergoes phase-slippage events, where the local phase difference at the barrier exceeds π [see Figs. 5(b) and 5(c)]. These events result in the creation of vortex-antivortex pairs near the junction, providing a dissipation channel for the injected pump energy. When averaged over many realizations, these stochastic slippage events give rise to dephasing in the imbalance dynamics, as shown in Fig. 5(d). This dephasing reflects the cumulative impact of vortex formation, which gradually suppresses coherent amplification and ultimately sets the limit for the achievable gain.

To understand the underlying nonlinear mechanism of the JPA, we examine the time evolution of the density and the phase at the junction in Fig. 6. Without pump, the barrier velocity modulation results in a smooth excitation of the density, which is created by the change of junction phase that is synchronized with the signal frequency. When the pump field is on, there are higher-order density excitations that result from the modulation of the phase by the pump field. These can be explained with a perturbative solution of the equation $\tilde{I}_1 \cos(\omega_s t) = (1 + A_d \sin(\omega_d t)) \sin \phi$, assuming a small modulation parameter A_d and expanding ϕ as $\phi = \phi_0 + A_d \phi_1 + \mathcal{O}(A_d^2)$. At zeroth order, we obtain $\phi_0 \approx \tilde{I}_1 \cos(\omega_s t)$, which explains the smooth oscillation of the phase without the pump. The first-order term gives $\phi_1 \approx -\tilde{I}_1 \cos(\omega_s t) \sin(\omega_d t)$, highlighting the intrinsic nonlinear mixing between the pump and signal field. In Fig. 6(d), the perturbative solutions consistently capture the dynamics of our atomic JPA.

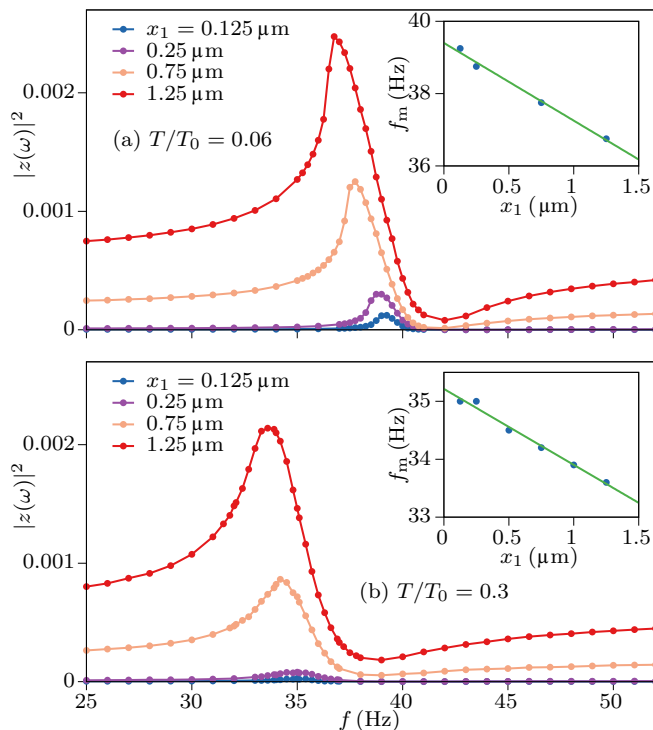


FIG. 7. (a), (b) Power spectrum of the imbalance, $|z(\omega)|^2$, for different values of the modulation amplitude x_1 , at $T/T_0 = 0.06$ and 0.3 . Inset shows the peak values of the spectrum as a function of x_1 , where the linear fit (continuous line) in the limit of $x_1 \rightarrow 0$ yields the resonance frequency $f_0 = 39.4$ and 35.2 Hz for $T/T_0 = 0.06$ and 0.3 , respectively.

Conclusions and outlook. We have studied the dynamics of a driven atomic JJ using classical-field simulations. The JJ is formed by coupling two two-dimensional atom clouds with a tunnel barrier, where we periodically modulate both the position (input signal) and the height (pump) of the barrier. By tuning the pump frequency to twice the Josephson plasma frequency, we have demonstrated parametric amplification for the signal. Our system works in the hertz frequency range, which is beyond the operating range of the superconducting JPAs, thereby offering potential for low-frequency signal detection and precision sensing of slowly varying fields, such as gravitational or magnetic fields [69–71]. The current cold-atom technology allows to accurately image the microscopic dynamics of the system [60,61]. In this case, we have an unprecedented access to the density and phase *microscopic* dynamics at the junction responsible for the amplification and the intrinsic nonlinear mixing between the signal and pump fields. Our results can be experimentally probed using *in situ* imaging and matter-wave interferometry, where the former enables measurement of the density imbalance and the latter allows detection of the underlying JPA mechanism. Our proposed protocol can be implemented in driven Josephson junctions experimentally realized in Refs. [60,61].

Our findings have broad implications for the understanding of the basic mechanism on which parametric amplification relies in quantum systems. For the specific case of quantum electronics devices, while the microscopic phase-density

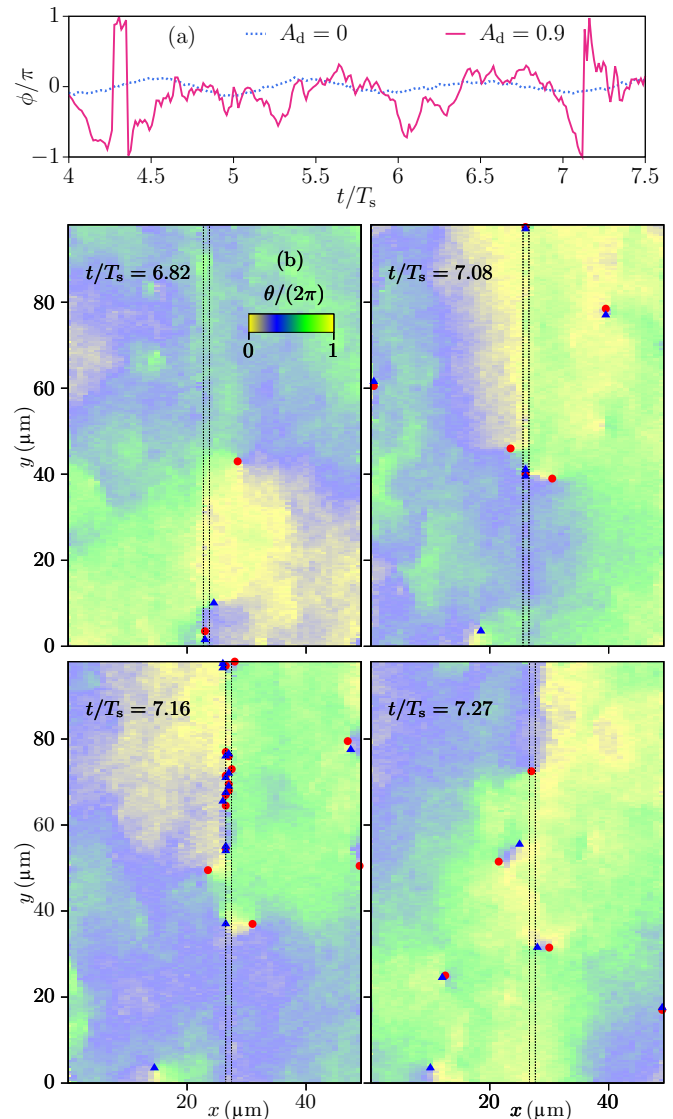


FIG. 8. (a) Junction phase $\phi(t)$ for a single sample at $\tilde{I}_1 = 0.42$ with (continuous line) and without pump (dashed line), during the time interval between 4 and 7.5 cycles. T_s is the signal oscillation period. (b) Snapshots of the phase distribution $\theta(x, y)$ at times near seventh cycle, for the same sample and parameters as in panel (a). The vertical dashed lines denote the location of the driven barrier. Vortices (triangles) and antivortices (dots) are identified based on the local phase winding; see text.

dynamics can be inferred through suitable Josephson current measurements, their direct experimental probing is out of the reach of current techniques, such as scanning probe microscopy.

Acknowledgments. L.M. acknowledges funding by the Deutsche Forschungsgemeinschaft (DFG) in the excellence cluster “Advanced Imaging of Matter”—EXC 2056—project ID 390715994 and by ERDF of the European Union and “Fonds of the Hamburg Ministry of Science, Research, Equalities and Districts (BWFGB).”

Data availability. The data that support the findings of this Letter are not publicly available. The data are available from the authors upon reasonable request.

Appendix A: Josephson plasma frequency. The protocol with periodic modulation of the barrier position results in coherent oscillations of the current across the junction, as demonstrated by the time evolution of the imbalance $z(t)$ across the junction in the main text. Since the system behaves as a driven harmonic oscillator, a maximum response is expected at the resonant frequency of the junction. To identify this resonance, we calculate the power spectrum of the imbalance, $|z(\omega)|^2$, using a numerical Fourier transform. In Fig. 7, we show $|z(\omega)|^2$ determined for various values of the modulation amplitude x_1 at temperatures of $T/T_0 = 0.06$ and 0.3 . The spectrum features a resonance peak, which we track at varying x_1 . Using a linear scaling approach, we determine the resonance frequency f_0 in the limit of vanishing x_1 , which gives $f_0 = 39.4$ and 35.2 Hz for $T/T_0 = 0.06$ and 0.3 , respectively.

Appendix B: Dephasing dynamics. Here, we elaborate on the dephasing dynamics observed in the density imbalance at

large pump amplitudes A_d . During time evolution, the junction phase exhibits strong fluctuations that lead to the formation of vortex-antivortex pairs near the junction. This is confirmed by analyzing the phase distribution in individual samples across the ensemble. Figure 8 shows the time evolution of the phase field $\theta(x, y)$ during the interval in which the junction phase undergoes a strong change. Vortex excitations are identified by computing the phase winding around the lattice plaquette of size $l \times l$ using $\sum_{\square} \delta\theta(x, y) = \delta_x\theta(x, y) + \delta_y\theta(x + l, y) + \delta_x\theta(x + l, y + l) + \delta_y\theta(x, y + l)$, where $\theta(x, y)$ is the phase field of $\psi(x, y)$ and the phase differences between sites are taken to be $\delta_{x/y}\theta(x, y) \in (-\pi, \pi]$. We identify a vortex (antivortex) by a phase winding of 2π (-2π). The total number of vortices N_v is obtained by counting all vortices and antivortices in a given sample. This number is then averaged over the initial ensemble and the time evolution between 5 and 15 cycles.

-
- [1] W. B. Case, The pumping of a swing from the standing position, *Am. J. Phys.* **64**, 215 (1996).
 - [2] T.G. van de Roer, *Microwave Electronic Devices*, Microwave Technology Series (Chapman & Hall, United Kingdom, 1994).
 - [3] B. R. Mollow and R. J. Glauber, Quantum theory of parametric amplification. I, *Phys. Rev.* **160**, 1076 (1967).
 - [4] R. Baumgartner and R. Byer, Optical parametric amplification, *IEEE J. Quantum Electron.* **15**, 432 (1979).
 - [5] I. Thomson and G. Janssens, High reliability parametric amplifier varactor diodes for satellite applications, *Microelectron. Reliab.* **15**, 563 (1976).
 - [6] B. Yurke, Squeezed-state generation using a Josephson parametric amplifier, *J. Opt. Soc. Am. B* **4**, 1551 (1987).
 - [7] B. Yurke, L. R. Corruccini, P. G. Kaminsky, L. W. Rupp, A. D. Smith, A. H. Silver, R. W. Simon, and E. A. Whittaker, Observation of parametric amplification and deamplification in a Josephson parametric amplifier, *Phys. Rev. A* **39**, 2519 (1989).
 - [8] I. Siddiqi, R. Vijay, F. Pierre, C. M. Wilson, M. Metcalfe, C. Rigetti, L. Frunzio, and M. H. Devoret, RF-driven Josephson bifurcation amplifier for quantum measurement, *Phys. Rev. Lett.* **93**, 207002 (2004).
 - [9] E. A. Tholén, A. Ergül, E. M. Doherty, F. M. Weber, F. Grégis, and D. B. Haviland, Nonlinearities and parametric amplification in superconducting coplanar waveguide resonators, *Appl. Phys. Lett.* **90**, 253509 (2007).
 - [10] T. Yamamoto, K. Inomata, M. Watanabe, K. Matsuba, T. Miyazaki, W. D. Oliver, Y. Nakamura, and J. S. Tsai, Flux-driven Josephson parametric amplifier, *Appl. Phys. Lett.* **93**, 042510 (2008).
 - [11] Y. Asaki, B. Alcalde Pampliega, P. G. Edwards, S. Iguchi, and E. J. Murphy, Astronomical radio interferometry, *Nat. Rev. Methods Primers* **3**, 89 (2023).
 - [12] M. E. Marhic, P. A. Andrekson, P. Petropoulos, S. Radic, C. Peucheret, and M. Jazayerifar, Fiber optical parametric amplifiers in optical communication systems, *Laser Photon. Rev.* **9**, 50 (2015).
 - [13] S. V. Dvornikov, A. F. Kryachko, I. A. Velmisov, and D. A. Zatuchny, *Amplifiers in Radio Receivers: Characteristics, Operating Principles, and Efficiency*, Springer Aerospace Technology (Springer, 2022).
 - [14] M. Koch, D. M. Mittleman, J. Ornik, and E. Castro-Camus, Terahertz time-domain spectroscopy, *Nat. Rev. Methods Primers* **3**, 48 (2023).
 - [15] N. Du *et al.* (ADMX Collaboration), Search for invisible axion dark matter with the axion dark matter experiment, *Phys. Rev. Lett.* **120**, 151301 (2018).
 - [16] A. M. Zagoskin, E. Il'ichev, M. W. McCutcheon, J. F. Young, and F. Nori, Controlled generation of squeezed States of microwave radiation in a superconducting resonant circuit, *Phys. Rev. Lett.* **101**, 253602 (2008).
 - [17] Z. Y. Ou, X. Y. Zou, L. J. Wang, and L. Mandel, Observation of nonlocal interference in separated photon channels, *Phys. Rev. Lett.* **65**, 321 (1990).
 - [18] E. M. Nagasako, S. J. Bentley, R. W. Boyd, and G. S. Agarwal, Nonclassical two-photon interferometry and lithography with high-gain parametric amplifiers, *Phys. Rev. A* **64**, 043802 (2001).
 - [19] M. A. Castellanos-Beltran, K. D. Irwin, G. C. Hilton, L. R. Vale, and K. W. Lehnert, Amplification and squeezing of quantum noise with a tunable Josephson metamaterial, *Nat. Phys.* **4**, 929 (2008).
 - [20] A. Blais, A. L. Grimsmo, S. M. Girvin, and A. Wallraff, Circuit quantum electrodynamics, *Rev. Mod. Phys.* **93**, 025005 (2021).
 - [21] C. L. Degen, F. Reinhard, and P. Cappellaro, Quantum sensing, *Rev. Mod. Phys.* **89**, 035002 (2017).
 - [22] S. D. Bass and M. Doser, Quantum sensing for particle physics, *Nat. Rev. Phys.* **6**, 329 (2024).
 - [23] C. Macklin, K. O'Brien, D. Hover, M. E. Schwartz, V. Bolkhovskiy, X. Zhang, W. D. Oliver, and I. Siddiqi, A near-quantum-limited Josephson traveling-wave parametric amplifier, *Science* **350**, 307 (2015).
 - [24] L. Amico, A. Osterloh, and F. Cataliotti, Quantum many particle systems in ring-shaped optical lattices, *Phys. Rev. Lett.* **95**, 063201 (2005).
 - [25] L. Amico, *et al.*, Roadmap on atomtronics: State of the art and perspective, *AVS Quantum Sci.* **3**, 039201 (2021).
 - [26] L. Amico, D. Anderson, M. Boshier, J.-P. Brantut, L.-C. Kwek, A. Minguzzi, and W. von Klitzing, Colloquium: Atomtronic circuits: From many-body physics to quantum technologies, *Rev. Mod. Phys.* **94**, 041001 (2022).

- [27] A. Ramanathan, K. C. Wright, S. R. Muniz, M. Zelan, W. T. Hill, C. J. Lobb, K. Helmerson, W. D. Phillips, and G. K. Campbell, Superflow in a toroidal Bose-Einstein condensate: An atom circuit with a tunable weak link, *Phys. Rev. Lett.* **106**, 130401 (2011).
- [28] S. Eckel, J. G. Lee, F. Jendrzejewski, N. Murray, C. W. Clark, C. J. Lobb, W. D. Phillips, M. Edwards, and G. K. Campbell, Hysteresis in a quantized superfluid ‘atomtronic’ circuit, *Nature (London)* **506**, 200 (2014).
- [29] C. Ryu, P. W. Blackburn, A. A. Blinova, and M. G. Boshier, Experimental realization of Josephson junctions for an atom SQUID, *Phys. Rev. Lett.* **111**, 205301 (2013).
- [30] C. Ryu, E. C. Samson, and M. G. Boshier, Quantum interference of currents in an atomtronic SQUID, *Nat. Commun.* **11**, 3338 (2020).
- [31] C.-C. Chien, S. Peotta, and M. Di Ventra, Quantum transport in ultracold atoms, *Nat. Phys.* **11**, 998 (2015).
- [32] S. Krinner, T. Esslinger, and J.-P. Brantut, Two-terminal transport measurements with cold atoms, *J. Phys.: Condens. Matter* **29**, 343003 (2017).
- [33] J. Polo, W. J. Chetcuti, E. C. Domanti, P. Kitson, A. Osterloh, F. Perciavalle, V. P. Singh, and L. Amico, Perspective on new implementations of atomtronic circuits, *Quantum Sci. Technol.* **9**, 030501 (2024).
- [34] F. S. Cataliotti, S. Burger, C. Fort, P. Maddaloni, F. Minardi, A. Trombettoni, A. Smerzi, and M. Inguscio, Josephson junction arrays with Bose-Einstein condensates, *Science* **293**, 843 (2001).
- [35] M. Albiez, R. Gati, J. Fölling, S. Hunsmann, M. Cristiani, and M. K. Oberthaler, Direct observation of tunneling and nonlinear self-trapping in a single Bosonic Josephson junction, *Phys. Rev. Lett.* **95**, 010402 (2005).
- [36] L. J. LeBlanc, A. B. Bardou, J. McKeever, M. H. T. Extavour, D. Jervis, J. H. Thywissen, F. Piazza, and A. Smerzi, Dynamics of a tunable superfluid junction, *Phys. Rev. Lett.* **106**, 025302 (2011).
- [37] G. Spagnolli, G. Semeghini, L. Masi, G. Ferioli, A. Trenkwalder, S. Coop, M. Landini, L. Pezzè, G. Modugno, M. Inguscio, A. Smerzi, and M. Fattori, Crossing over from attractive to repulsive interactions in a tunneling Bosonic Josephson junction, *Phys. Rev. Lett.* **118**, 230403 (2017).
- [38] M. Pigneur, T. Berrada, M. Bonneau, T. Schumm, E. Demler, and J. Schmiedmayer, Relaxation to a phase-locked equilibrium state in a one-dimensional Bosonic Josephson junction, *Phys. Rev. Lett.* **120**, 173601 (2018).
- [39] S. Levy, E. Lahoud, I. Shomroni, and J. Steinhauer, The a.c. and d.c. Josephson effects in a Bose-Einstein condensate, *Nature (London)* **449**, 579 (2007).
- [40] W. J. Kwon, G. D. Pace, R. Panza, M. Inguscio, W. Zwerger, M. Zaccanti, F. Scazza, and G. Roati, Strongly correlated superfluid order parameters from dc Josephson supercurrents, *Science* **369**, 84 (2020).
- [41] G. Del Pace, W. J. Kwon, M. Zaccanti, G. Roati, and F. Scazza, Tunneling transport of unitary Fermions across the superfluid transition, *Phys. Rev. Lett.* **126**, 055301 (2021).
- [42] V. P. Singh, E. Bernhart, M. Röhrle, H. Ott, L. Mathey, and L. Amico, Weak-link to tunneling regime in a 3D atomic Josephson junction, *arXiv:2509.03591*.
- [43] K. S. Gan, V. P. Singh, L. Amico, and R. Dumke, Josephson dynamics in 2D ring-shaped condensates, *arXiv:2509.00533*.
- [44] N. Luick, L. Sobirey, M. Bohlen, V. P. Singh, L. Mathey, T. Lompe, and H. Moritz, An ideal Josephson junction in an ultra-cold two-dimensional Fermi gas, *Science* **369**, 89 (2020).
- [45] E. Rydow, V. P. Singh, A. Beregi, E. Chang, L. Mathey, C. J. Foot, and S. Sunami, Observation of a bilayer superfluid with interlayer coherence, *Nat. Commun.* **16**, 7201 (2025).
- [46] K. C. Wright, R. B. Blakestad, C. J. Lobb, W. D. Phillips, and G. K. Campbell, Driving phase slips in a superfluid atom circuit with a rotating weak link, *Phys. Rev. Lett.* **110**, 025302 (2013).
- [47] K. Görg, L. Mathey, and V. P. Singh, Realizing an atomic quantum interference device in a rotating-box potential, *Phys. Rev. Res.* **7**, 043340 (2025).
- [48] A. Smerzi, S. Fantoni, S. Giovanazzi, and S. R. Shenoy, Quantum coherent atomic tunneling between two trapped Bose-Einstein condensates, *Phys. Rev. Lett.* **79**, 4950 (1997).
- [49] S. Raghavan, A. Smerzi, S. Fantoni, and S. R. Shenoy, Coherent oscillations between two weakly coupled Bose-Einstein condensates: Josephson effects, π oscillations, and macroscopic quantum self-trapping, *Phys. Rev. A* **59**, 620 (1999).
- [50] S. Giovanazzi, A. Smerzi, and S. Fantoni, Josephson effects in dilute Bose-Einstein condensates, *Phys. Rev. Lett.* **84**, 4521 (2000).
- [51] S. Jährling, V. P. Singh, and L. Mathey, Designing atomtronic circuits via superfluid dynamics, *Phys. Rev. Res.* **7**, 033112 (2025).
- [52] B. Zhu, V. P. Singh, J. Okamoto, and L. Mathey, Dynamical control of the conductivity of an atomic Josephson junction, *Phys. Rev. Res.* **3**, 013111 (2021).
- [53] S. Kohler and F. Sols, Chemical potential standard for atomic Bose-Einstein condensates, *New J. Phys.* **5**, 94 (2003).
- [54] A. Eckardt, T. Jinasundera, C. Weiss, and M. Holthaus, Analog of photon-assisted tunneling in a Bose-Einstein condensate, *Phys. Rev. Lett.* **95**, 200401 (2005).
- [55] J. Grond, T. Betz, U. Hohenester, N. J. Mauser, J. Schmiedmayer, and T. Schumm, The Shapiro effect in atomchip-based bosonic Josephson junctions, *New J. Phys.* **13**, 065026 (2011).
- [56] N. Pradhan, R. Kanamoto, M. Bhattacharya, and P. K. Mishra, Fractional Shapiro steps in a cavity-coupled Josephson ring condensate, *Phys. Rev. Res.* **7**, 043188 (2025).
- [57] V. P. Singh, J. Polo, L. Mathey, and L. Amico, Shapiro steps in driven atomic Josephson junctions, *Phys. Rev. Lett.* **133**, 093401 (2024).
- [58] S. Shapiro, Josephson currents in superconducting tunneling: The effect of microwaves and other observations, *Phys. Rev. Lett.* **11**, 80 (1963).
- [59] C. C. Grimes and S. Shapiro, Millimeter-wave mixing with Josephson junctions, *Phys. Rev.* **169**, 397 (1968).
- [60] G. D. Pace, D. Hernández-Rajkov, V. P. Singh, N. Grani, M. F. Fernández, G. Nesti, J. A. Seman, M. Inguscio, L. Amico, and G. Roati, Shapiro steps in strongly-interacting Fermi gases, *Science* **390**, 1125 (2025).
- [61] E. Bernhart, M. Röhrle, V. P. Singh, L. Mathey, L. Amico, and H. Ott, Observation of Shapiro steps in an ultracold atomic Josephson junction, *Science* **390**, 1130 (2025).
- [62] P. Blakie, A. Bradley, M. Davis, R. Ballagh, and C. Gardiner, Dynamics and statistical mechanics of ultra-cold Bose gases using c-field techniques, *Adv. Phys.* **57**, 363 (2008).
- [63] A. Polkovnikov, Phase space representation of quantum dynamics, *Ann. Phys. (NY)* **325**, 1790 (2010).

- [64] V. P. Singh, W. Weimer, K. Morgener, J. Siegl, K. Hueck, N. Luick, H. Moritz, and L. Mathey, Probing superfluidity of Bose-Einstein condensates via laser stirring, *Phys. Rev. A* **93**, 023634 (2016).
- [65] V. P. Singh and L. Mathey, Sound propagation in a two-dimensional Bose gas across the superfluid transition, *Phys. Rev. Res.* **2**, 023336 (2020).
- [66] N. Prokof'ev, O. Ruebenacker, and B. Svistunov, Critical point of a weakly interacting two-dimensional Bose gas, *Phys. Rev. Lett.* **87**, 270402 (2001).
- [67] N. Prokof'ev and B. Svistunov, Two-dimensional weakly interacting Bose gas in the fluctuation region, *Phys. Rev. A* **66**, 043608 (2002).
- [68] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/7s1n-ddp9> for the driven junction spectrum, the current-chemical potential characteristic, and the driven RSJ circuit model, which includes Refs. [57,72].
- [69] D. Sheng, A. R. Perry, S. P. Krzyzewski, S. Geller, J. Kitching, and S. Knappe, A microfabricated optically-pumped magnetic gradiometer, *Appl. Phys. Lett.* **110**, 031106 (2017).
- [70] J. Javor, Z. Yao, M. Imboden, D. K. Campbell, and D. J. Bishop, Analysis of a Casimir-driven parametric amplifier with resilience to Casimir pull-in for MEMS single-point magnetic gradiometry, *Microsyst. Nanoeng.* **7**, 73 (2021).
- [71] K. Somiya, K. Suzuki, S. Otabe, and K. Harada, Intracavity signal amplification system for next-generation gravitational-wave detectors, *Phys. Rev. D* **107**, 122005 (2023).
- [72] V. P. Singh, N. Luick, L. Sobirey, and L. Mathey, Josephson junction dynamics in a two-dimensional ultracold Bose gas, *Phys. Rev. Res.* **2**, 033298 (2020).